



Unit 4 · Outcome 2 · Energy Sources

Fossil fuel use & peak oil

KK04

VCE Environmental Science · Units 3 & 4 · Exam study summary

"Changes in the rate of fossil fuel use over time and the concept of peak oil."

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



A1 Three fossil-fuel eras

For almost all of human history, our energy came from muscle, wood, wind and water. Fossil fuels changed that — but they did not all arrive at once. They scaled up in sequence, each unlocked by a new technology.

From the Industrial Revolution (~1760s onward), coal powered steam engines, iron-making and railways. It was the first fossil fuel to scale, and it dominated global energy into the early 20th century.

After the first commercial well (1859) and the internal combustion engine, oil surged through the 20th century. Cars, trucks, ships, aviation and petrochemicals made oil the backbone of modern transport.

Gas grew strongly from the mid-1900s for heating, industry and electricity. It is the fastest-growing fossil fuel today, often framed as a lower-carbon "bridge" fuel — though it is still a fossil fuel.

Add them together and the global picture is a roughly exponential climb through the 1900s. But notice the shape carefully: total use is still rising, while the rate of growth is slowing, and coal use has plateaued (and may have peaked) in many regions as gas and renewables displace it. "The rate of fossil fuel use over time" is not a single number — it is a curve with a story.

A2 What pushes the rate up — and back down

If a question asks you to explain changes in the rate, you need causes, not just a description of the line. Examiners want the drivers on both sides.

- Population growth — more people, more energy demanded
- Industrialisation & economic growth — factories, construction, freight
- Rising living standards — appliances, cars, air-con, travel
- Transport & electrification spreading worldwide
- Cheap, energy-dense, easy to store — fossil fuels were simply convenient
- Energy efficiency — doing more with each unit of fuel
- Renewables & substitution — solar, wind and gas displacing coal
- Policy & regulation — carbon targets, emissions limits, EV uptake
- Price — high prices curb demand and reward alternatives
- Resource depletion — the easy reserves get used first

◆ ANALOGY

Analogy — a city's traffic The rate of fossil fuel use is like cars on a freeway. For a century, more lanes kept opening (population, industry, transport) and traffic only grew. Lately, working-from-home, trains and electric cars (efficiency, substitution, policy) have started easing the flow in wealthy nations — even though, globally, the freeway is still busy.

B1 Peak oil: Hubbert and the curve

Now the second half. "Peak oil" is a precise idea — and the single most-tested thing about it is what it does not mean.

The point in time at which the maximum rate of oil extraction (production) is reached, after which the rate of production enters a terminal, irreversible decline.

In 1956, geoscientist M. King Hubbert reasoned that because oil is a finite, non-renewable resource, its production over time must trace a roughly symmetrical bell-shaped curve — now called a Hubbert curve. Production rises as fields are found and developed, reaches a peak, then falls as reserves deplete. Hubbert predicted that conventional oil production in the United States would peak around 1970 — and it did, almost exactly.

⚠ WATCH OUT

The trap that costs the most marks Peak oil does NOT mean oil is "running out." At the peak, roughly half of the recoverable oil still remains underground. What changes is the rate you can pull it out. And because the cheapest, easiest, highest-quality oil is taken first, the remaining half is harder, slower and more expensive to extract.

💠 ANALOGY

Analogy — the lolly jar at a party Picture a big jar of lollies. Everyone grabs the easy ones off the top first, so the grab rate (lollies per minute) climbs fast. It peaks while the jar is still about half full — then slows, because people now have to dig awkwardly for the ones stuck at the bottom. Peak oil is the peak of the grab rate, not the empty jar. The lollies aren't gone; they're just harder to reach.

B2 Why the global peak keeps moving

Hubbert nailed the US peak, but global "peak oil" has been predicted — and missed — many times. Knowing why turns a 2-mark recall into a 4-mark explanation.

Factor	How it pushes the peak later (or reshapes the curve)
New discoveries	Fields once thought not to exist (deepwater, Brazil's pre-salt) added recoverable oil, lifting and stretching the curve.
Better extraction technology	Horizontal drilling and hydraulic fracturing unlocked US shale / tight oil from ~2010, transforming the supply picture and delaying the supposed peak.
Unconventional sources	Oil sands, tight oil, coal seam gas and oil shale count as oil-equivalent supply once technology and prices allow.
Price	High prices make difficult, low-grade fields economic, so reserves that were "uneconomic" suddenly count.
Demand	Efficiency, EVs and renewables can lower demand — which, as we'll see, can bring the peak forward from the other side.

✓ TIP

The deeper cost: EROEI Even when technology delays the peak, it rarely makes oil cheaper in energy terms. EROEI (energy return on energy invested) falls as we chase harder oil — you must burn more energy to win each barrel. A gusher of easy crude might return 100:1; tar sands might return closer to 5:1. Falling EROEI is a quieter, more reliable signal than any single "peak year."

B3 Peak supply vs peak demand

The original peak-oil idea was a supply story: production peaks because the resource is finite. But the conversation has flipped. Many analysts now argue the peak may arrive because demand falls away first —

driven by electric vehicles, efficiency, renewables and climate policy — rather than because we can't pump any more.

◆ ANALOGY

Analogy — why did the party end? Did the music stop because the band ran out of songs (peak supply) or because the crowd went home (peak demand)? Both empty the dancefloor — but for opposite reasons. Modern peak-oil debate is mostly about which one happens first.

As of 2026 the timing is genuinely contested: some major outlooks see oil demand reaching a plateau later this decade as EV sales and efficiency bite, while others project demand still growing for years yet. For the exam, the safe, sophisticated line is: peak oil is no longer just about supply running short — it is increasingly about demand falling away, and credible forecasts disagree on the timing. Showing you know the debate exists is worth more than betting on one date.

◆ NOTE

Why this matters for energy choices If you believe oil will peak on the supply side, you prepare for scarcity and price shocks. If you believe it peaks on the demand side, you prepare for stranded assets — expensive oil infrastructure built just as the world stops needing it. Either way, the policy conclusion points the same direction: build the transition early.

★ Bass Strait: a Hubbert curve on Victoria's doorstep

You don't need to look overseas for peak oil. Off the Gippsland coast — the view from our case-study town, Tarra Cove — the real Bass Strait fields tell the whole story in miniature.

The Gippsland Basin (Bass Strait), run by Esso/ExxonMobil with Woodside, has produced oil and gas for more than 50 years. Its oil output rose, peaked and is now in mature decline — about half of all the wells no longer produce, and decommissioning of the spent platforms begins in 2027. A textbook Hubbert curve, right here.

As local crude declined, Australia came to import roughly 80–90% of its liquid fuel. Two refineries closed in 2021 (Kwinana in WA and Altona in Victoria), leaving just two nationally — Viva Energy in Geelong and Ampol's Lytton in Brisbane. That raises real fuel-security questions when global supply is disrupted.

✓ TIP

Use this in the exam If a question asks for a real example of peak oil or of changing fossil-fuel use, Bass Strait / the Gippsland Basin is a high-value Australian case: a maturing field in clear decline, driving a national shift from local production to imported fuel, with platform rehabilitation now underway (which links forward to KK08).

02 Decode the command word

◆ COMMAND WORDS

The verb tells you what shape your answer must take. Tap one to see what KK04 expects.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

03 Six exam traps

⚠ EXAM TRAPS

Every one of these has cost real students real marks. Read them as a pre-flight checklist.

⚠ WATCH OUT

Trap 1 — "Peak oil means we're running out" The single biggest error. Peak oil is the peak of the extraction rate; about half the resource remains at the peak. Say rate, not "amount left".

⚠ WATCH OUT

Trap 2 — Describing the curve instead of explaining it "Explain" needs causes. Don't just say "use went up then levelled off" — name drivers (industrialisation, transport, then efficiency, renewables, policy).

⚠ WATCH OUT

Trap 3 — Treating "fossil fuels" as one thing Coal, oil and gas have different time-trajectories. Coal led and is plateauing; gas is still growing. Distinguish them where the marks allow.

⚠ WATCH OUT

Trap 4 — Forgetting why predictions kept changing If asked why the global peak shifted, you must mention discoveries, technology (fracking/shale), unconventional sources and price — not just "they were wrong".

⚠ WATCH OUT

Trap 5 — Missing the supply-vs-demand flip Top answers note that peak oil may now be driven by falling demand (EVs, efficiency, policy), not only by limited supply.

⚠ WATCH OUT

Trap 6 — No real example A named case (Bass Strait decline, US 1970 peak, or the US shale surge) lifts a generic answer into a high band. Keep one ready.

MARKS

04 P·E·L trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

Build a high-band answer in three moves — Point, Evidence, Link. Type into each box; the trainer flags the key ideas examiners look for.

05 Same question, two answers

Q: "Define peak oil and explain why it does not mean the world is about to run out of oil." (3 marks)

"Peak oil is when oil starts to run out and there isn't much left. It doesn't mean we run out straight away because there's still some oil in other countries we can use."

"Peak oil is the point at which the maximum rate of oil extraction is reached, after which production permanently declines. It does not mean oil is running out, because at the peak roughly half the recoverable resource still remains underground. What declines is the rate of production, as the cheap, easy oil is used first and the remainder is harder and costlier to extract."

06 Practice exam questions

◆ NOTE

Five VCAA-style questions · 20 marks total Write your answer, reveal the marking guide, then self-mark. Your score saves automatically and feeds the dock.

- Improved extraction technology — horizontal drilling and hydraulic fracturing unlocked shale/tight oil that earlier models assumed was unrecoverable, adding supply.
- New discoveries — fields such as deepwater and pre-salt reserves added recoverable oil not counted before.
- Unconventional sources — oil sands and tight oil became viable, expanding the resource base.
- Higher prices — made previously uneconomic, harder-to-reach fields worth extracting.
- Define / frame: peak oil = peak of the extraction rate; traditionally a supply -side idea (finite resource → Hubbert curve).
- Supply still matters: oil is finite, EROEI falls as easy oil is used up, and supply shocks still drive price spikes — so a supply-led peak is not impossible.
- Demand-side case: EVs, efficiency, renewables and climate policy are slowing the rate of oil use; several major outlooks now see demand plateauing later this decade, suggesting a demand-led peak.
- Uncertainty: credible forecasts disagree on timing — some still project growth — so any single date is contestable.
- Judgement: a defensible conclusion (e.g. that the balance has shifted toward demand in wealthy economies, while supply constraints still bite globally), explicitly supported by the points above.

Fossil fuel use & peak oil

"Changes in the rate of fossil fuel use over time and the concept of peak oil."

Overview

This dot point has two halves, and the exam rewards you for handling both cleanly. First, how the rate of fossil fuel use has changed over time — the long arc from coal to oil to gas, and why that arc is now bending. Second, the concept of peak oil — a specific idea about the rate of extraction from a finite resource, which is almost always tested through one killer misconception.

✓ TIP

The one-line version Use over time: fossil fuel use grew explosively across the 20th century and is now plateauing in places. Peak oil: the moment the rate of oil extraction tops out and begins a permanent decline — not the moment oil runs out.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Peak oil is when oil starts to run out and there isn't much left. It doesn't mean we run out straight away because there's still some oil in other countries we can use." Why it loses marks: the definition is wrong — it equates peak oil with running out, the exact trap. No mention of extraction rate, no "half remains" idea, and the second sentence is vague geography rather than the real reason.

✓ STRONG / MODEL ANSWER

Peak oil is the point at which the maximum rate of oil extraction is reached, after which production permanently declines. It does not mean oil is running out, because at the peak roughly half the recoverable resource still remains underground. What declines is the rate of production, as the cheap, easy oil is used first and the remainder is harder and costlier to extract." Why it scores: precise definition built on rate; directly tackles the misconception; uses the half-remains fact; explains the mechanism (easy oil first). Three clean, separable marking points.